

Freight Rate Prediction — Machine Learning Engineer Assessment

Spotter Labs | Assessment Report

1. Executive Summary

This submission develops a freight-rate regression model using the supplied labeled development data and applies it to the 12,000-load validation set and the fixed December prediction scenario. The workflow covers data profiling, quality assessment, feature engineering, time-aware validation, model comparison, hyperparameter refinement, final training, prediction generation, and structural validation with the provided scorer.

The selected model is a HistGradientBoostingRegressor using absolute-error loss, with `learning_rate=0.05`, `max_iter=300`, `max_leaf_nodes=31`, and `l2_regularization=5.0`. Model selection used expanding-window monthly validation, with MAE as the primary metric and RMSE as a secondary metric.

2. Data Overview

The labeled development dataset contains 48,000 rows covering January 1 through October 31, 2025. The supplied validation dataset contains 12,000 rows covering November 1 through December 31, 2025. The fixed December input contains 31 rows, one for each day of December 2025.

Dataset	Rows	Columns	Date coverage
Development	48,000	14	Jan–Oct 2025
Validation	12,000	13	Nov–Dec 2025
December inputs	31	7	Dec 2025

3. Data Exploration and Quality Findings

The development data had no duplicate rows and no duplicate `load_id` values. Distance was complete and positive, while `posted_rate` was complete and positive with a strong right-skew and a long upper tail. The target was therefore not clipped or winsorized solely because of extreme values.

Weight contained 300 missing values and 292 negative values. Negative weights were retained rather than silently converted because their target-rate distribution was similar to the positive-weight population and the records were spread across equipment types and months. Missing weight values were passed through the model's native handling.

`market_index` contained 374 missing development values. `quote_signal` was complete, but both `market_index` and `quote_signal` were excluded from the final common feature pipeline because the fixed December input does not provide those fields and the assessment does not specify a reconstruction method for them.

The validation data contains eight pickup/delivery cities not seen in the development categorical levels. It also contains 736 unseen route combinations, representing 1,461 validation loads. Geographic coordinates were therefore retained as numerical features to provide generalisation information beyond exact route identity.

4. Feature Engineering

The final common feature set contains 14 model inputs: `pickup`, `delivery`, `equipment`, `distance`, `weight`, `day_of_week`, `day_of_year`, `day_of_year_sin`, `day_of_year_cos`, `days_since_start`, `pickup_lat`, `pickup_lon`, `delivery_lat`, and `delivery_lon`.

Date features capture recurring weekly and annual structure. Sine/cosine transforms represent annual cyclicity. Geographic coordinates provide continuous location information. `load_id` is treated only as an identifier and is not used as a predictor.

5. Validation Strategy

A random train/test split was not used because the task is explicitly temporal: the model is trained on January–October development data and then applied to later November–December loads. Instead, model selection used expanding-window, one-month-ahead validation folds.

Fold	Training period	Validation month
1	Jan–Jun 2025	Jul 2025
2	Jan–Jul 2025	Aug 2025
3	Jan–Aug 2025	Sep 2025
4	Jan–Sep 2025	Oct 2025

MAE was used as the primary selection metric because it is directly interpretable in the target's rate units. RMSE was retained as a secondary diagnostic because it gives additional visibility into large errors.

All preprocessing and feature transformations within each fold were fitted using only the corresponding training portion. The validation target was not used during feature preparation or tuning. November–December data remained untouched during model selection and was used only for the final prediction stage.

6. Model Experiments

Candidate	Model / change	Mean MAE	Mean RMSE
Baseline	Monthly median-rate baseline	1138.41	1550.11
1	HistGradientBoosting, common features	167.03	644.21
2	+ geographic coordinates	161.71	645.48
2B	+ weight quality indicators	166.41	643.50
3	ExtraTreesRegressor	163.47	698.07
4	HGB with log1p target	126.92	629.42
5	HGB absolute-error loss	117.95	627.51
5A	Lower LR / more iterations	117.13	627.38
5B	More leaf nodes	117.10	627.65
5C	Higher L2 regularization	116.85	627.39

Candidate 5C was selected because it produced the lowest mean MAE across the four time-aware folds. The improvement over the nearby hyperparameter variants was modest, so experimentation was stopped rather than extended into an unnecessarily large search.

7. Final Model

HistGradientBoostingRegressor

Loss: absolute_error

Learning rate: 0.05

Maximum iterations: 300

Maximum leaf nodes: 31

L2 regularization: 5.0

Random state: 42

The final model was retrained on all 48,000 development rows from January through October 2025 before generating the untouched validation predictions and the fixed December predictions.

8. Final Prediction and Scorer Validation

The completed validation_predictions.csv contains exactly 12,000 rows with the required load_id and predicted_rate columns. Predictions are finite and positive, with no missing values or duplicate IDs.

The fixed December scenario was also generated for all 31 days. The route is Lexington to Fort Wayne, distance 360 miles, Dry Van equipment, and 32,000 lb weight; only the date changes. Coordinates for the two cities were reconstructed from the deterministic city-to-coordinate mapping observed in the development data.

The provided scorer was executed successfully. It validated 12,000 final predictions, validated 31 fixed December predictions, and created the candidate December chart. The scorer itself does not calculate the final validation metrics; those are calculated by Spotter after submission.

9. Fixed December Prediction Chart

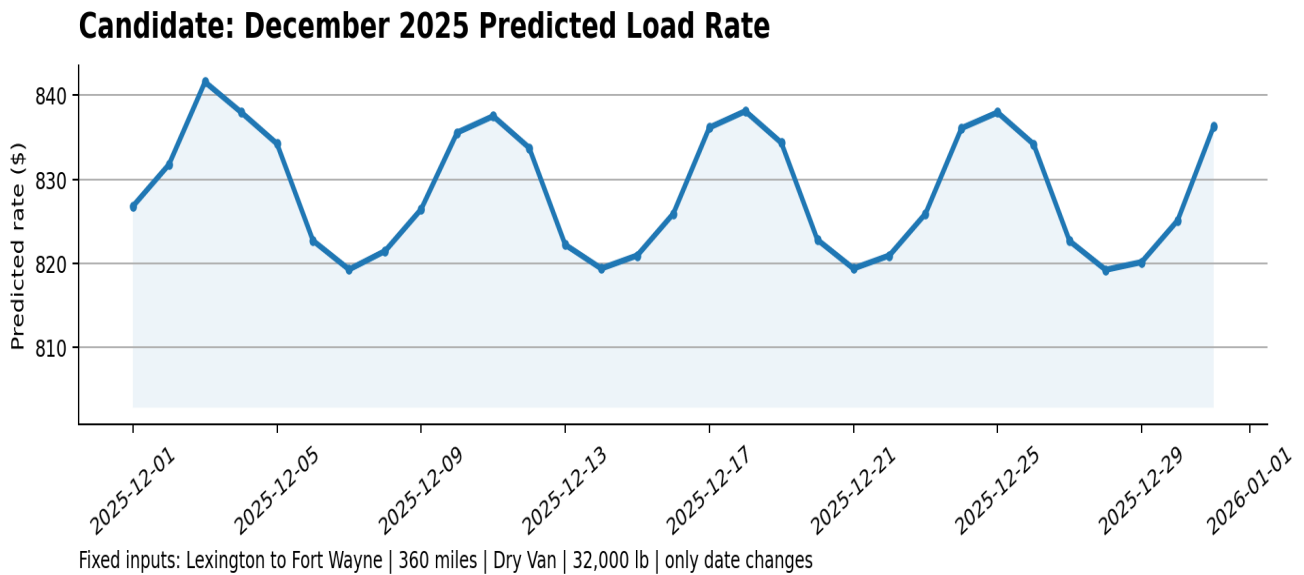


Figure 1. Candidate December 2025 predicted load rate for the fixed assessment scenario. The chart was generated from the completed December prediction file using the supplied scoring script.

10. Reproducibility and Submission Artifacts

The repository contains the supplied datasets, two Jupyter notebooks documenting profiling and model development, the reusable model implementation in `src/freight_rate_model.py`, `score.py`, `requirements.txt`, `validation_predictions.csv`, `december_predictions.csv`, the generated December chart, and this report.

The workflow is reproducible from the repository using the documented Python environment and dependencies. The notebooks provide the analysis trail, while the source module contains the final feature construction, model definition, training, and prediction functions.

End of report